

Ex.15

a) $A(x) = \frac{x}{x^2-4} = \frac{x}{(x+2)(x-2)}$ définie sur $\mathbb{R} \setminus \{-2; 2\}$

Soit $(a, b) \in \mathbb{R}^2$, $\forall x \in \mathbb{R} \setminus \{-2, 2\}$, $A(x) = \frac{a}{x+2} + \frac{b}{x-2}$ ①

1^{ère} méthode, par identification:

$$\textcircled{1} \Leftrightarrow \frac{x}{(x+2)(x-2)} = \frac{a(x-2) + b(x+2)}{(x+2)(x-2)} = \frac{(a+b)x + (-2a+2b)}{(x+2)(x-2)}$$

$$\Leftrightarrow \begin{cases} a+b=1 \\ -2a+2b=0 \end{cases} \Leftrightarrow \begin{cases} a=\frac{1}{2} \\ b=\frac{1}{2} \end{cases}$$

2^{ème} méthode, par calcul de limites:

$$\bullet a = \lim_{x \rightarrow -2} A(x)(x+2) = \lim_{x \rightarrow -2} \frac{x}{x-2} = \frac{-2}{-4} = \frac{1}{2}$$

$$\bullet b = \lim_{x \rightarrow 2} A(x)(x-2) = \lim_{x \rightarrow 2} \frac{x}{x+2} = \frac{2}{4} = \frac{1}{2}$$

donc $A(x) = \frac{1}{2(x+2)} + \frac{1}{2(x-2)} \quad \forall x \in \mathbb{R} \setminus \{-2, 2\}$.

b) $B(x) = \frac{2x+5}{(x-1)(x+2)(x-3)} \quad \forall x \in \mathbb{R} \setminus \{-2, 1, 3\}$

Soit $(a, b, c) \in \mathbb{R}^3$, $\forall x \in \mathbb{R} \setminus \{-2, 1, 3\}$,

$$B(x) = \frac{a}{x-1} + \frac{b}{x+2} + \frac{c}{x-3}$$

- $a = \lim_{x \rightarrow 1} B(x)(x-1) = \lim_{x \rightarrow 1} \frac{2x+5}{(x+2)(x-3)} = \frac{7}{3 \times (-2)} = -\frac{7}{6}$
- $b = \lim_{x \rightarrow -2} B(x)(x+2) = \frac{1}{15}$
- $c = \lim_{x \rightarrow 3} B(x)(x-3) = \frac{11}{10}$

donc $\forall x \in \mathbb{R} \setminus \{-2, 1, 3\}$,

$$B(x) = -\frac{7}{6} \times \frac{1}{x-1} + \frac{1}{15} \times \frac{1}{x+2} + \frac{11}{10} \times \frac{1}{x-3}$$

c) $C(x) = \frac{x^3 - 3x^2 + x - 4}{x-1} \quad \forall x \in \mathbb{R} \setminus \{1\}$

$\deg(\text{Num}) \geq \deg(\text{Den})$ donc division euclidienne:

$$\begin{array}{r|l} x^3 - 3x^2 + x - 4 & x-1 \\ -x^3 + x^2 & x^2 - 2x - 1 \\ \hline -2x^2 + x - 4 & \\ + 2x^2 - 2x & \\ \hline -x - 4 & \\ + x - 1 & \\ \hline -5 & \end{array}$$

donc $\forall x \in \mathbb{R} \setminus \{1\}$, $C(x) = x^2 - 2x - 1 - \frac{5}{x-1}$

d) $D(x) = \frac{2x^3 + x^2 - x + 1}{x^2 - 2x + 1} = \frac{2x^3 + x^2 - x + 1}{(x-1)^2} \quad \forall x \in \mathbb{R} \setminus \{1\}$

$\deg(\text{Num}) > \deg(\text{Den})$ donc division euclidienne.

$$\begin{array}{r|l} 2x^3 + x^2 - x + 1 & x^2 - 2x + 1 \\ -2x^3 + 4x^2 - 2x & 2x + 5 \\ \hline 5x^2 - 3x + 1 & \\ -5x^2 + 10x - 5 & \\ \hline 7x - 4 & \end{array}$$

$$\text{donc } \forall x \in \mathbb{R} \setminus \{1\}, D(x) = 2x + 5 + \frac{7x - 4}{(x-1)^2}$$

racine double

1^{ère} méthode: Identification

$$\exists (a, b) \in \mathbb{R}^2, \forall x \in \mathbb{R} \setminus \{1\}, \overbrace{\frac{7x-4}{(x-1)^2}}^{D_1(x)} = \frac{a}{x-1} + \frac{b}{(x-1)^2}$$

$$\Leftrightarrow \frac{7x-4}{(x-1)^2} = \frac{a(x-1)+b}{(x-1)^2} = \frac{ax + (-a+b)}{(x-1)^2}$$

$$\Leftrightarrow \begin{cases} a=7 \\ -a+b=-4 \end{cases} \Leftrightarrow \underline{\underline{\begin{cases} a=7 \\ b=3 \end{cases}}}$$

2^{ème} méthode: Calcul de limites

$$\bullet b = \lim_{x \rightarrow 1} D_1(x) (x-1)^2 = \lim_{x \rightarrow 1} 7x - 4 = 3$$

$$\bullet a = \lim_{x \rightarrow 1} (D_1(x) (x-1)^2)' = \lim_{x \rightarrow 1} (7x - 4)' = 7$$

$$\text{donc } \forall x \in \mathbb{R} \setminus \{1\}, D(x) = 2x + 5 + D_1(x)$$

$$\underline{\underline{D(x) = 2x + 5 + \frac{7}{x-1} + \frac{3}{(x-1)^2}}}$$

$$e) E(x) = \frac{x^5 + x^4 + 1}{x^3 - x} = \frac{x^5 + x^4 + 1}{x(x+1)(x-1)} \quad \forall x \in \mathbb{R} \setminus \{-1, 0, 1\}$$

$\deg(\text{Num}) \geq \deg(\text{Den})$: division euclidienne

$$\begin{array}{r|l} x^5 + x^4 + 1 & x^3 - x \\ -x^5 + x^3 & x^2 + x + 1 \\ \hline x^4 + x^3 + 1 & \\ -x^4 + x^2 & \\ \hline x^3 + x^2 + 1 & \\ -x^3 + x & \\ \hline x^2 + x + 1 & \end{array}$$

$$\text{donc } \forall x \in \mathbb{R} \setminus \{-1, 0, 1\}, E(x) = x^2 + x + 1 + \overbrace{\frac{x^2 + x + 1}{x^3 - 1}}^{E_1(x)}$$

Soit $(a, b, c) \in \mathbb{R}^3$, $\forall x \in \mathbb{R} \setminus \{-1, 0, 1\}$,

$$E_1(x) = \frac{a}{x} + \frac{b}{x+1} + \frac{c}{x-1}$$

$$\bullet a = \lim_{x \rightarrow 0} E_1(x) \times x = \lim_{x \rightarrow 0} \frac{x^2 + x + 1}{(x+1)(x-1)} = -1$$

$$\bullet b = \lim_{x \rightarrow -1} E_1(x) \times (x+1) = \frac{1}{2}$$

$$\bullet c = \lim_{x \rightarrow 1} E_1(x) \times (x-1) = \frac{3}{2}$$

$$\text{donc } \forall x \in \mathbb{R} \setminus \{-1, 0, 1\}, E(x) = x^2 + x + 1 - \frac{1}{x} + \frac{1}{2(x+1)} + \frac{3}{2(x-1)}$$

$$f) F(x) = \frac{x^3 + x + 1}{(x-1)^3(x+1)} \quad \forall x \in \mathbb{R} \setminus \{-1, 1\}$$

(3)

Soit $(a, b, c, d) \in \mathbb{R}^4$, $\forall x \in \mathbb{R} \setminus \{-1, 1\}$,

$$F(x) = \frac{a}{x-1} + \frac{b}{(x-1)^2} + \frac{c}{(x-1)^3} + \frac{d}{x+1}$$

• On peut comme d'habitude procéder par identification, mais trop long ici.

$$c = \lim_{x \rightarrow 1} F(x) \times (x-1)^3 = \lim_{x \rightarrow 1} \frac{x^3 + x + 1}{x+1} = \underline{\underline{\frac{3}{2}}}$$

$$d = \lim_{x \rightarrow -1} F(x)(x+1) = \lim_{x \rightarrow -1} \frac{x^3 + x + 1}{(x-1)^3} = \frac{-1}{-8} = \underline{\underline{\frac{1}{8}}}$$

$$\begin{aligned} b &= \lim_{x \rightarrow 1} (F(x)(x-1)^3)' = \lim_{x \rightarrow 1} \left(\frac{x^3 + x + 1}{x+1} \right)' \\ &= \lim_{x \rightarrow 1} \frac{(3x^2 + 1)(x+1) - (x^3 + x + 1)}{(x+1)^2} \\ &= \underline{\underline{\frac{4 \times 2 - 3}{2^2}}} \end{aligned}$$

$$\underline{\underline{b = \frac{5}{4}}}$$

• pour a on utilise la dérivée seconde de $F(x)(x-1)^3$, mais $\Delta \left(a(x-1)^2 + b(x-1) + c + \frac{d(x-1)^3}{x+1} \right)'' = 2a + \left(\frac{d(x-1)^3}{x+1} \right)''$

$$\text{ainsi } 2a = \lim_{x \rightarrow 1} (F(x)(x-1)^3)'' = \lim_{x \rightarrow 1} \left(\frac{2x^3 + 3x^2}{(x+1)^2} \right)'$$

$$= \lim_{x \rightarrow 1} \frac{(6x^2 + 6x)(x+1)^2 - (2x+2)(2x^3 + 3x^2)}{(x+1)^4}$$

$$2a = \underline{\underline{\frac{7}{4}}}$$

$$\underline{\underline{(\Rightarrow a = \frac{7}{8}}}$$

Autre méthode pour trouver a:

$$F(x) = \frac{\text{Num}(x)}{\text{Den}(x)}$$

$$\begin{cases} \deg(\text{Num}(x)) = 3 \\ \deg(\text{Den}(x)) = 4 \end{cases} \quad \text{et } 4-3=1,$$

→ on étudie la limite de $F(x) \times x$ quand $x \rightarrow +\infty$

$$\lim_{x \rightarrow +\infty} F(x) \times x = \lim_{x \rightarrow +\infty} \frac{ax}{x-1} + \frac{bx}{(x-1)^2} + \frac{cx}{(x-1)^3} + \frac{dx}{x+1}$$

$$\Rightarrow \lim_{x \rightarrow +\infty} \frac{x(x^3+x+1)}{(x-1)^3(x+1)} = a + 0 + 0 + d$$

$$\Rightarrow a + d = 1 \quad \text{donc } a = 1 - d = \frac{7}{8},$$

donc $\forall x \in \mathbb{R} \setminus \{-1, 1\}$,

$$F(x) = \frac{7}{8} \times \frac{1}{x-1} + \frac{5}{4} \times \frac{1}{(x-1)^2} + \frac{3}{2} \times \frac{1}{(x-1)^3} + \frac{1}{8} \times \frac{1}{x+1}$$

$$9) G(x) = \frac{x+1}{x^4+1} \quad \forall x \in \mathbb{R},$$

On a montré à l'ex. 13 c) que $\forall x \in \mathbb{R}$,
 $x^4+1 = (x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)$

$$\text{donc } \forall x \in \mathbb{R}, G(x) = \frac{x+1}{\underbrace{(x^2+\sqrt{2}x+1)}_{\text{polynôme du second degré}} \underbrace{(x^2-\sqrt{2}x+1)}_{\text{polynôme du second degré}}}$$

polynômes du second degré
ne s'annulant pas sur $\mathbb{R}$

→ décomposition de 2^{de} espèce.

donc $\exists (a, b, c, d) \in \mathbb{R}^4$, $\forall x \in \mathbb{R}$,

$$G(x) = \frac{ax+b}{x^2+\sqrt{2}x+1} + \frac{cx+d}{x^2-\sqrt{2}x+1}$$

1^{ère} méthode: par identification

$$G(x) = \frac{(ax+b)(x^2-\sqrt{2}x+1) + (cx+d)(x^2+\sqrt{2}x+1)}{x^4+1}$$

$$G(x) = \frac{(a+c)x^3 + (-\sqrt{2}a + \sqrt{2}c + b+d)x^2 + (a+c - \sqrt{2}b + \sqrt{2}d)x + b+d}{x^4+1}$$

$$\Leftrightarrow \begin{cases} a+c=0 \\ b+d=1 \\ -\sqrt{2}a + \sqrt{2}c + b+d=0 \\ a+c - \sqrt{2}b + \sqrt{2}d = 1 \end{cases}$$

$$\Leftrightarrow \begin{cases} a=-c \\ b=1-d \\ 2\sqrt{2}c = -1 \\ d\sqrt{2}(2) - \sqrt{2} = 1 \end{cases} \quad \Leftrightarrow \begin{cases} a = \frac{\sqrt{2}}{4} \\ b = \frac{1}{2} - \frac{\sqrt{2}}{4} \\ c = -\frac{\sqrt{2}}{4} \\ d = \frac{1}{2} + \frac{\sqrt{2}}{4} \end{cases}$$

2^{ème} méthode: par calcul de limites et valeurs remarquables:

$$\bullet \lim_{x \rightarrow +\infty} G(x) \times x = \lim_{x \rightarrow +\infty} \frac{ax^2+bx}{x^2+\sqrt{2}x+1} + \lim_{x \rightarrow +\infty} \frac{cx^2+dx}{x^2+\sqrt{2}x+1}$$

$$\Leftrightarrow \lim_{x \rightarrow +\infty} \frac{x^2+x}{x^4+1} = a+c$$

$$\Leftrightarrow \underline{a = -c}$$

$$• G(0) = \frac{ax+0+b}{0^2+\sqrt{2}x+1} + \frac{cx+0+d}{0^2-\sqrt{2}x+1}$$

$$\Leftrightarrow \frac{1}{1} = b+d \Leftrightarrow \underline{b = 1-d},$$

$$• G(1) = \frac{a+b}{1+\sqrt{2}+1} + \frac{c+d}{1-\sqrt{2}+1} \Leftrightarrow \frac{2}{2} = \dots$$

→ 4 conditions permettant de trouver a, b, c et d.

$$\text{Ainsi } \forall x \in \mathbb{R}, G(x) = \frac{\frac{\sqrt{2}}{4}x + \left(\frac{1}{2} - \frac{\sqrt{2}}{4}\right)}{x^2 + \sqrt{2}x + 1} + \frac{-\frac{\sqrt{2}}{4}x + \left(\frac{1}{2} + \frac{\sqrt{2}}{4}\right)}{x^2 - \sqrt{2}x + 1}$$

$$h) H(x) = \frac{x}{(x^2+1)(x^2+4)} \quad \forall x \in \mathbb{R}$$

polynômes du second degré
ne s'annulant pas sur $\mathbb{R}$.

$$\forall x \in \mathbb{R}, \exists (a, b, c, d) \quad H(x) = \frac{ax+b}{x^2+1} + \frac{cx+d}{x^2+4}$$

$$\Leftrightarrow \frac{x}{(x^2+1)(x^2+4)} = \frac{(a+c)x^3 + (b+d)x^2 + (4a+c)x + 4b+d}{(x^2+1)(x^2+4)}$$

$$\rightarrow \text{on identifie:} \quad \Leftrightarrow \begin{cases} a+c=0 \\ b+d=0 \\ 4a+c=1 \\ 4b+d=0 \end{cases} \Leftrightarrow \begin{cases} a = \frac{1}{3} \\ b=0 \\ c = -\frac{1}{3} \\ d=0 \end{cases}$$

$$\text{donc } \forall x \in \mathbb{R}, H(x) = \frac{1}{3} \frac{x}{x^2+1} - \frac{1}{3} \frac{x}{x^2+4}$$

$$i) I(x) = \frac{2x^4 + x^3 + 3x^2 - 6x + 1}{2x^3 - x^2} \quad \forall x \in \mathbb{R} \setminus \{0; \frac{1}{2}\}$$

$$\deg(\text{Num}) > \deg(\text{Den})$$

$$\begin{array}{r|l} 2x^4 + x^3 + 3x^2 - 6x + 1 & 2x^3 - x^2 \\ - 2x^4 + x^3 & \\ \hline 2x^3 + 3x^2 - 6x + 1 & \\ - 2x^3 + x^2 & \\ \hline 4x^2 - 6x + 1 & \end{array}$$

$$\text{donc } \forall x \in \mathbb{R} \setminus \{0; \frac{1}{2}\}, I(x) = x + 1 + \overbrace{\frac{4x^2 - 6x + 1}{x^2(2x-1)}}^{I_1(x)}$$

$$\text{donc } \exists (a, b, c) \in \mathbb{R}^3, \forall x \in \mathbb{R} \setminus \{0; \frac{1}{2}\},$$

$$I(x) = x + 1 + \frac{a}{x} + \frac{b}{x^2} + \frac{c}{2x-1}$$

$$\bullet c = \lim_{x \rightarrow \frac{1}{2}} I_1(x) \times (2x-1) = \lim_{x \rightarrow \frac{1}{2}} \frac{4x^2 - 6x + 1}{x^2} = -4$$

$$\bullet b = \lim_{x \rightarrow 0} I_1(x) \times (x^2) = -1$$

$$\bullet \lim_{x \rightarrow +\infty} I_1(x) \times x = \lim_{x \rightarrow +\infty} \frac{4x^2 - 6x + 1}{2x^2 - x} = 2$$

$$\text{et } \lim_{x \rightarrow +\infty} \left(\frac{a}{x} + \frac{b}{x^2} + \frac{c}{2x-1} \right) \times x = a + \frac{c}{2}$$

$$\text{donc } a + \frac{c}{2} = 2 \quad (\Rightarrow) \quad a = 4$$

$$\text{donc } \forall x \in \mathbb{R} \setminus \{0; \frac{1}{2}\}, I(x) = x + 1 + \frac{4}{x} - \frac{1}{x^2} - \frac{4}{2x-1}$$

Ex. 16

$$1) I(x) = \frac{(3-2i)x - 5 + 3i}{x^2 + ix + 2} \quad \forall x \in D_I \quad \text{ensemble à définir}$$

$$\text{or } x^2 + ix + 2 = 0 : \Delta = -1 - 8 = -9$$

$$\sqrt{\Delta} = 3i \text{ ou } -3i \quad (\text{on ne garde qu'une des 2 racines!})$$

$$x_1 = \frac{-i - 3i}{2} = -2i$$

$$x_2 = \frac{-i + 3i}{2} = i$$

$$\text{donc } D_I = \mathbb{C} \setminus \{-2i, i\}$$

$$\text{et } \exists (a, b) \in \mathbb{C}^2, \forall x \in D_I,$$

$$I(x) = \frac{a}{x+2i} + \frac{b}{x-i}$$

$$\Leftrightarrow \frac{(3-2i)x - 5 + 3i}{x^2 + ix + 2} = \frac{(a+b)x + (-ai + 2bi)}{x^2 + ix + 2}$$

$$\rightarrow \text{on identifie:} \quad \Leftrightarrow \begin{cases} a+b = 3-2i \\ -ai + 2bi = -5 + 3i \end{cases}$$

$$\Leftrightarrow \begin{cases} a = 3-2i-b \\ -(3-2i-b)i + 2bi = -5 + 3i \end{cases}$$

$$\Leftrightarrow \begin{cases} 3bi - 2 - 3i = -5 + 3i \\ a = 3-2i-b \end{cases}$$

$$\Leftrightarrow \begin{cases} 3bi = -3 + 6i \\ a = 3-2i-b \end{cases} \Leftrightarrow \begin{cases} b = 2+i \\ a = 1-3i \end{cases}$$

$$\text{donc } \forall x \in D_I,$$

$$I(x) = \frac{1-3i}{x+2i} + \frac{2+i}{x-i}$$

$$J(x) = \frac{x+i}{x^2+i}$$

$$\forall x \in \mathbb{C} \setminus \overbrace{\{e^{-i\frac{\pi}{4}}; e^{i\frac{3\pi}{4}}\}}^{D_J}$$

$$\cos x^2 + i = 0 \Leftrightarrow x^2 = -i$$

$$\Leftrightarrow x^2 = e^{-i(\frac{\pi}{2} + 2k\pi)} \quad (k \in \mathbb{Z})$$

$$\Leftrightarrow x = e^{-i(\frac{\pi}{4} + k\pi)}$$

$$\Leftrightarrow x \in \left\{ e^{-i\frac{\pi}{4}}; e^{i\frac{3\pi}{4}} \right\}$$

on rappelle que :

$$\begin{cases} e^{-i\frac{\pi}{4}} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \\ e^{i\frac{3\pi}{4}} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \end{cases}$$

donc $\exists (a, b) \in \mathbb{C}^2, \forall x \in D_J,$

$$J(x) = \frac{a}{x - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i} + \frac{b}{x + \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i}$$

$$\Leftrightarrow \frac{x+i}{x^2+i} = \frac{(a+b)x + (\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i)a + (-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i)b}{x^2+1}$$

→ on identifie :

$$\begin{cases} a+b=1 \\ (\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i)a + (-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i)b = i \end{cases}$$

$$\Leftrightarrow \begin{cases} a=1-b \\ \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i - \frac{\sqrt{2}}{2}b + \frac{\sqrt{2}}{2}ib - \frac{\sqrt{2}}{2}b + \frac{\sqrt{2}}{2}ib = i \end{cases}$$

$$\Leftrightarrow \begin{cases} b(-\sqrt{2} + \sqrt{2}i) = -\frac{\sqrt{2}}{2} + (1 + \frac{\sqrt{2}}{2})i \\ a=1-b \end{cases}$$

$$\Leftrightarrow \begin{cases} b = \frac{-\frac{\sqrt{2}}{2} + (1 + \frac{\sqrt{2}}{2})i}{-\sqrt{2} + \sqrt{2}i} = \frac{(-\frac{\sqrt{2}}{2} + (1 + \frac{\sqrt{2}}{2})i)(-\sqrt{2} - \sqrt{2}i)}{2+2} \\ a=1-b \end{cases}$$

$$= \frac{2 + \sqrt{2}}{4} - \frac{\sqrt{2}}{4}i$$

$$\Rightarrow \begin{cases} a = \frac{2-\sqrt{2}}{4} + \frac{\sqrt{2}}{4}i \\ b = \frac{2+\sqrt{2}}{4} - \frac{\sqrt{2}}{4}i \end{cases}$$

$$\text{donc } \forall x \in D_J, J(x) = \frac{\frac{2-\sqrt{2}}{4} + \frac{\sqrt{2}}{4}i}{x - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i} + \frac{\frac{2+\sqrt{2}}{4} - \frac{\sqrt{2}}{4}i}{x + \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i}$$

$$k(x) = \frac{2x}{(x+i)^2} \quad \forall x \in \mathbb{C} \setminus \{i\}$$

$$\exists (a, b) \in \mathbb{C}, \quad \forall x \in \mathbb{C} \setminus \{i\},$$

$$k(x) = \frac{a}{x+i} + \frac{b}{(x+i)^2}$$

$$\Rightarrow \frac{2x}{(x+i)^2} = \frac{a(x+i) + b}{(x+i)^2}$$

$$\Rightarrow \begin{cases} a = 2 \\ a + b = 0 \end{cases} \Rightarrow \begin{cases} a = 2 \\ b = -2i \end{cases}$$

$$\text{donc } \forall x \in \mathbb{C} \setminus \{i\},$$

$$k(x) = \frac{2}{x+i} - \frac{2i}{(x+i)^2}$$

$$2) L(x) = \frac{x^2-3}{(x^2+1)(x^2+4)} \quad \forall x \in \mathbb{R}$$

• dans $\mathbb{R}$:

polynômes du second degré sans racine réelle.

$$\exists (a, b, c, d) \in \mathbb{R}^4, \quad \forall x \in \mathbb{R}, \quad L(x) = \frac{ax+b}{x^2+1} + \frac{cx+d}{x^2+4}$$

$$\Rightarrow \frac{x^2-3}{(x^2+1)(x^2+4)} = \frac{(a+c)x^3 + (b+d)x^2 + (4a+c)x + 4b+d}{(x^2+1)(x^2+4)}$$

On identifie: $\Leftrightarrow \begin{cases} a+c=0 \\ b+d=1 \\ 4a+c=0 \\ 4b+d=-3 \end{cases} \Leftrightarrow \begin{cases} a=0 \\ b=-\frac{4}{3} \\ c=0 \\ d=\frac{7}{3} \end{cases}$

donc $\forall x \in \mathbb{R}$,

$$L(x) = -\frac{4}{3}x \frac{1}{x^2+1} + \frac{7}{3} \frac{1}{x^2+4}$$

• dans $\mathbb{C}$:

• $x^2+1=0 \Leftrightarrow x \in \{-i; i\}$

• $x^2+4=0 \Leftrightarrow x \in \{-2i; 2i\}$

donc $\exists (a, b, c, d) \in \mathbb{C}^4$, $\forall x \in \mathbb{C} \setminus \{-2i; 2i; -i; i\}$,

$$L(x) = \frac{a}{x+i} + \frac{b}{x-i} + \frac{c}{x+2i} + \frac{d}{x-2i}$$

• $a = \lim_{x \rightarrow -i} L(x)(x+i) = \lim_{x \rightarrow -i} \frac{x^2-3}{(x-i)(x^2+4)} = -\frac{2}{3}i$

• $b = \lim_{x \rightarrow i} L(x)(x-i) = \frac{2}{3}i$

• $c = \lim_{x \rightarrow -2i} L(x)(x+2i) = \frac{7}{12}i$

• $d = \lim_{x \rightarrow 2i} L(x)(x-2i) = -\frac{7}{12}i$

donc $\forall x \in \mathbb{C} \setminus \{-2i; 2i; -i; i\}$

$$L(x) = \frac{-2i}{3(x+i)} + \frac{2i}{3(x-i)} + \frac{7i}{12(x+2i)} - \frac{7i}{12(x-2i)}$$

$$H(x) = \frac{x^5 + x + 1}{x^4 - 1} \quad \forall x \in \mathbb{R} \setminus \{-1; 1\}$$

dans $\mathbb{R}$: $\deg(\text{Num}) \geq \deg(\text{Den})$

$$\begin{array}{r|l} x^5 + x + 1 & x^4 - 1 \\ -x^5 + x & x \\ \hline 2x + 1 & \end{array}$$

donc $\forall x \in \mathbb{R} \setminus \{-1; 1\}$,

$$H(x) = x + \frac{2x+1}{x^4-1} \quad M_1(x)$$

$$\text{or } \forall x \in \mathbb{R}, x^4 - 1 = \underbrace{(x^2+1)(x^2-1)}_{\text{polynôme de degré 2 sans racine réelle}} = (x^2+1)(x+1)(x-1)$$

polynôme de degré 2
sans racine réelle

donc $\exists (a, b, c, d) \in \mathbb{R}^4$,

$$\forall x \in \mathbb{R} \setminus \{-1; 1\},$$

$$H(x) = x + \underbrace{\frac{ax+b}{x^2+1} + \frac{c}{x+1} + \frac{d}{x-1}}_{M_1(x)}$$

$$\bullet c = \lim_{x \rightarrow -1} M_1(x)(x+1)$$

$$= \lim_{x \rightarrow -1} \frac{2x+1}{(x^2+1)(x-1)} = \underline{\underline{\frac{1}{4}}}$$

$$\bullet d = \lim_{x \rightarrow 1} M_1(x)(x-1) = \underline{\underline{\frac{3}{4}}}$$

$$\bullet \lim_{x \rightarrow +\infty} M_1(x)x = \lim_{x \rightarrow +\infty} \frac{ax^2+bx}{x^2+1} + \frac{cx}{x+1} + \frac{dx}{x-1} = a+c+d$$

$$\text{or } \lim_{x \rightarrow +\infty} M_1(x)x = \lim_{x \rightarrow +\infty} \frac{2x^2+x}{x^4-1} = 0$$

$$\text{donc } a = -c - d = \underline{\underline{-1}}$$

$$\bullet M(0) = \frac{ax_0+b}{0^2+1} + \frac{c}{0+1} + \frac{d}{0-1} = b+c-d$$

$$\text{or } M(0) = \frac{2x_0+1}{0^4-1} = -1$$

$$\text{donc } b = -1 - c + d = \underline{-\frac{1}{2}}$$

donc $\forall x \in \mathbb{R} \setminus \{-1; 1\}$,

$$M(x) = x + \underbrace{\frac{-x - \frac{1}{2}}{x^2+1}}_{M_2(x)} + \frac{1}{4} \frac{1}{x+1} + \frac{3}{4} \frac{1}{x-1}$$

dans $\mathbb{C}$:

$$x^2+1=0 \Leftrightarrow x \in \{-i; i\}$$

donc $\exists (a, b) \in \mathbb{C}^2, \forall x \in \mathbb{C} \setminus \{-i; i\}$,

$$\begin{aligned} M_2(x) &= \frac{-x - \frac{1}{2}}{x^2+1} = \frac{a}{x+i} + \frac{b}{x-i} \\ &= \frac{(a+b)x - ai + bi}{x^2+1} \end{aligned}$$

$$\Leftrightarrow \begin{cases} a+b = -1 \\ -ai + bi = -\frac{1}{2} \end{cases}$$

$$\Leftrightarrow \begin{cases} a = -1-b \\ -(-1-b)i + bi = -\frac{1}{2} \end{cases}$$

$$\Leftrightarrow \begin{cases} 2bi = -\frac{1}{2} - i \\ a = -1-b \end{cases} \Leftrightarrow \begin{cases} a = -\frac{1}{2} - \frac{1}{4}i \\ b = -\frac{1}{2} + \frac{1}{4}i \end{cases}$$

donc $\forall x \in \mathbb{C} \setminus \{-i; i, -1; 1\}$,

$$M(x) = x + \frac{-\frac{1}{2} - \frac{1}{4}i}{x+i} + \frac{-\frac{1}{2} + \frac{1}{4}i}{x-i} + \frac{1}{4} \frac{1}{x+1} + \frac{3}{4} \frac{1}{x-1}$$